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Small Signal Modeling and Stability Analysis of N-Phase Interleaved Boost Converter

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Abstract— **Solar PVs are emerging as one of the better alternatives for power generation. The output from solar cells requires boost topologies for obtaining an adequate level of voltage. Interleaving techniques in boost topologies lead to a better current sharing and a reduction in current ripple. This paper presents a small signal model for n phase interleaved boost converter (IBC). The transfer functions are derived considering the parasitics of inductors, capacitor and power electronic switch. Such considerations result in accurate mathematical models closely pertaining to a practical IBC. The resulting analysis will ensure development of effective control strategies. The derived mathematical model's accuracy is investigated by carrying out open loop control studies in the frequency domain on a 3-phase IBC using MATLAB®/SIMULINK.**

Keywords—continuous conduction mode (CCM), interleaved boost converter,small signal analysis,lead lag compensator

I. INTRODUCTION

The global power outlook has witnessed a paradigm shift, as the focus of power generation has transitioned to renewable sources of energy. Solar PVs are gaining more attention than other renewable sources when electricity production is considered. The power output from these systems alone is not adequate to satisfy the consumers' need. Thus voltage stepping is a necessity to bring the power levels at par with the grid. Switched DC –DC converters have proven to be an efficient technology in renewable energy based power systems [1]-[4].

The output from solar PV lies in the range of 150-180 V. Boost converter, a switched DC-DC converter topology is extensively used for stepping up PV cell's low voltage output. These converters consist of non linear elements acting as switch along with capacitor, inductor and resistance. Thus boost converter can be categorized as a non linear time variant systems.

Linearization techniques are applied for modeling switched converters for using a linear controller with them.[5]

Boost converters consists of a single switch and hence to maximize the output, the designer has to rely on increasing size of inductor, resulting in more component losses, along with an increase in input current and output voltage ripple.[6] Nowadays interleaving techniques in boost converter are used with the desired motive of reduced current ripple and increased output power[7]. Modeling of a multi level interleaved boost converter (IBC) was done in the past using Lunze transformation [8], [9]. However, state space technique is preferred for most of the analysis due to its simpler approach [10]. Existing analysis and modeling is mostly based for a 2 phase IBC [11]-[13]. It becomes increasingly difficult to design a controller for an IBC as its phases are increased due to complex nature of equations and transfer functions. The renewable energy sector is increasingly focusing on IBC with 2 and 3 phase topologies dominating the scenario [14].

Small signal analysis is widely accepted linearization technique involving state space averaging and perturbation. An overview of previous work reveals the lack of any comprehensive model for N level IBC. This work addresses this issue and presents a small signal model. While deriving the transfer functions, the consideration of all the parasitic of circuital elements results in accurate mathematical models resembling to an actual IBC A lead-lag compensator is designed to test the stability of the open loop system. The variation of output voltage and inductor current with duty ratio is analyzed .The gain performance is simulated with the variation in duty ratio.

The paper is divided as follows: Section II and III focuses on development of state space model and final equations for 3-level IBC. Section IV deals with the stability analysis and checks the feasibility of derived transfer function models. Section V follows a discussion of obtained results. Finally the paper ends with a conclusion in Section VI.

II. OPERATIONAL DETAILS

The ideal circuit of the N-level IBC topology is shown in Fig. 1.

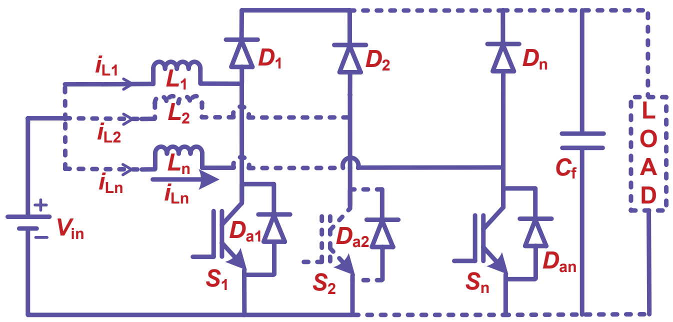

Fig. 1 Generalized circuit for N- Level ideal interleaved boost converter

The IBC consist of N inductors connected in parallel with N switches operating by switching signals with same phase shift 360/*N*; ensuring a regularized charging of inductors.
By considering the operation in CCM mode for d< 0.5

$$d_{2i-1} = d \qquad (1)$$

$$d_{2i} = \frac{1}{N} - d \qquad (2)$$

e.g. – For a 3 level IBC,

$$d_1 = d_3 = d_5 = d \;;$$

$$d_2 = d_4 = d_6 = 0.33 - d \;;$$

The sum of duty ratio is 0.33 .After first mode, the second switching signal will be applied $2\pi/3$ phase shifted. Thus no 2 inductors can charge simultaneously, resulting in less tedious analysis. For state space analysis, the preferred state variables are inductor current and capacitor voltage. The circuital analysis for every mode of operation is performed and averaging is done by taking the product with duty cycle of that particular mode and subsequent summation.

III. Small Signal Analysis

Assuming the state variables inductor current and capacitor voltage the state space equations can be defined as

$$\dot{x} = Ax + BV_{in} \qquad (3)$$

$$v = Cx \qquad (4)$$

Where,

$$A = \sum_{i=1}^{2N} d_i A_i \qquad (5)$$

$$B = \sum_{i=1}^{2N} d_i B_i \qquad (6)$$

$$C = \sum_{i=1}^{2N} d_i C_i \qquad (7)$$

Using the relation (1), (2) in (5), (6) and (7)

$$A = D\sum_{i=1}^{N} A_{2i-1} + \left(\frac{1}{N} - D\right)\sum_{i=1}^{N} A_{2i} \qquad (8)$$

$$B = D\sum_{i=1}^{N} B_{2i-1} + \left(\frac{1}{N} - D\right)\sum_{i=1}^{N} B_{2i} \qquad (9)$$

$$C = D\sum_{i=1}^{N} C_{2i-1} + \left(\frac{1}{N} - D\right)\sum_{i=1}^{N} C_{2i} \qquad (10)$$

From which the state space averaged equation for an N-level IBC can be obtained as

$$\frac{d}{dt}\begin{bmatrix} i_{L_1} \\ i_{L_2} \\ \vdots \\ i_{L_n} \\ v_{c_f} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1,n} & A_{1,n+1} \\ A_{21} & A_{22} & \cdots & A_{2,n} & A_{2,n+1} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ A_{1,n} & A_{2,n} & \cdots & A_{n,n} & A_{n,n+1} \\ A_{1,n+1} & A_{2,n+1} & \cdots & A_{n,n+1} & A_{n+1,n+1} \end{bmatrix} \begin{bmatrix} i_{L_1} \\ i_{L_2} \\ \vdots \\ i_{L_n} \\ v_c \end{bmatrix} + \begin{bmatrix} 1/L_1 \\ 1/L_2 \\ \vdots \\ 1/L_n \\ 0 \end{bmatrix} V_{in} \qquad (11)$$

$$v_o = \begin{bmatrix} C_{11} & C_{12} & \cdots & C_{1,n} & C_{1,n+1} \end{bmatrix} \begin{bmatrix} i_{L_1} \\ i_{L_2} \\ \vdots \\ i_{L_n} \\ v_{c_f} \end{bmatrix}$$

where,

$$A_{kk} = \left[\frac{-1}{L_k}\right]\left[(1-D)\left(r_{L_k} + r_{d_k} + \frac{Rr_c}{R + r_c}\right) + D\left(r_{L_k} + r_{s_k}\right)\right];$$

$$A_{k,n+1} = \left[\frac{-1}{L_k}\right]\left[(1-D)\left(\frac{R}{R + r_c}\right)\right];\; A_{n+1,k} = \left[(1-D)\left(\frac{R}{C(R + r_c)}\right)\right]$$

$$A_{n+1,n+1} = \frac{-1}{C(R + r_c)};$$

$$C_{1,n+1} = \frac{R}{R + r_c};\; C_{1,k} = \frac{(1-D)Rr_c}{R + r_c}$$

For $k \in [1, n]$,

Introducing perturbation in variables such that,

$$i_L = I_L + \hat{i}_L \;;\; v_c = V_c + \hat{v}_c \;;\; v_{in} = V_{in} + \hat{v}_{in}$$

$$v_o = V_o + \hat{v}_o \;;\; d = D + \hat{d}$$

After simplifying the Laplace transformation; considering the parameters in a new form as;

$$\gamma_k = sL_k + \left[(1-D)\left(r_{L_k} + r_{d_k} + \frac{Rr_c}{R + r_c}\right) + D\left(r_{L_k} + r_{s_k}\right)\right] \qquad (12)$$

$$\text{Let} \qquad r_{o_k} = r_{L_k} + r_{d_k} + \frac{Rr_c}{R + r_c} \qquad (13)$$

and, $$r_{\rho_k} = r_{L_k} + r_{s_k} \tag{14}$$

Thus $$\gamma_k = sL_k + (1-D)r_{\theta_k} + Dr_{\rho_k}; \tag{15}$$

Also $$\beta = \frac{(1-D)R}{R+r_c}; \tag{16}$$

$$\phi = \frac{(1-D)Rr_c}{R+r_c}; \tag{17}$$

$$\alpha = (-1)\left(sC_f + \frac{1}{R+r_c}\right); \tag{18}$$

$$\delta = \frac{R}{R+r_c}; \tag{19}$$

$$\eta_k = \left(r_{\theta_k} - r_{\rho_k}\right)\left(I_{L_k}\right) + V_c\frac{R}{R+r_c}; \tag{20}$$

The following perturbated equation is formed for n level IBC

$$\begin{bmatrix} \gamma_1 & 0 & \cdots & 0 & 0 & \beta \\ 0 & \gamma_2 & \cdots & 0 & 0 & \beta \\ \vdots & \vdots & \ddots & 0 & 0 & \beta \\ 0 & 0 & \cdots & \gamma_n & 0 & \beta \\ \beta & \beta & \cdots & \beta & 0 & -1 \\ \phi & \phi & \cdots & \phi & -1 & \delta \end{bmatrix} \begin{bmatrix} \hat{i}_{L_1}(s) \\ \hat{i}_{L_2}(s) \\ \vdots \\ \hat{i}_{L_n}(s) \\ \hat{v}_o(s) \\ \hat{v}_c(s) \end{bmatrix} = \begin{bmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_n \\ \delta(I_{L_1} + I_{L_2} \cdots I_{L_n}) \\ \frac{Rr_c}{R+r_c}(I_{L_1} + I_{L_2} \cdots I_{L_n}) \end{bmatrix} \hat{d}(s) + \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ 0 \\ 0 \end{bmatrix} \hat{v}_{in}(s) \tag{21}$$

Let a matrix M be defined such as,

$$M = \begin{bmatrix} \gamma_1 & 0 & \cdots & 0 & 0 & \beta \\ 0 & \gamma_2 & \cdots & 0 & 0 & \beta \\ \vdots & \vdots & \ddots & 0 & 0 & \beta \\ 0 & 0 & \cdots & \gamma_n & 0 & \beta \\ \beta & \beta & \cdots & \beta & 0 & -1 \\ \phi & \phi & \cdots & \phi & -1 & \delta \end{bmatrix}$$

The final perturbated eqn can be written as

$$\begin{bmatrix} \hat{i}_{L_1}(s) \\ \hat{i}_{L_2}(s) \\ \vdots \\ \hat{i}_{L_n}(s) \\ \hat{v}_o(s) \\ \hat{v}_{c_f}(s) \end{bmatrix} = M^{-1} \begin{bmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_n \\ \delta(I_{L_1} + I_{L_2} + \cdots I_{L_n}) \\ \frac{Rr_c}{R+r_c}(I_{L_1} + I_{L_2} + \cdots I_{L_n}) \end{bmatrix} \hat{d}(s) + M^{-1} \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ 0 \\ 0 \end{bmatrix} \hat{v}_{in}(s)$$

(22)

For N phase IBC, we obtain the transfer function of current in k^{th} Inductor leg/duty ratio, output voltage / duty ratio, current in 3^{rd} inductor leg/duty ratio, output voltage/duty ratio as in (23),(24), (25), (26) and voltage gain for the converter, taking into account the parasitic is given in (27). Thus for a 3 phase IBC, the final equations after small signal analysis are obtained by substituting $n = 3$.

$$\frac{\hat{i}_{L_k}(s)}{\hat{d}(s)} = \frac{1}{\alpha\prod_{i=1}^{k}\gamma_i - \beta^2\sum\prod_{i=1}^{k-1}\gamma_i}\left[\left\{\frac{\left(\prod_{i=1}^{k}\gamma_i\right)}{\gamma_k}\left(\beta^2\sum_{i=1}^{k-1}\frac{\eta_i-\eta_k}{\gamma_i} + \alpha\eta_k - \beta\delta\sum_{i=1}^{k} I_{L_i}\right)\right\}\right] \tag{23}$$

$$\frac{\hat{v}_o(s)}{\hat{d}(s)} = \frac{1}{\alpha\prod_{i=1}^{k}\gamma_i - \beta^2\sum\prod_{i=1}^{k-1}\gamma_i}\left\{(\alpha\phi - \beta\delta)\sum_{i=1}^{k}\frac{\eta_i}{\gamma_i} + \sum_{i=1}^{k} I_{L_i}\left(\delta - \frac{\alpha\phi}{1-D}\right)\right\}\prod_{i=1}^{k}\gamma_i \tag{24}$$

$$\frac{\hat{i}_{L_3}(s)}{\hat{d}(s)} = \frac{1}{\alpha\ (\gamma_1\gamma_2\gamma_3) - \beta^2(\gamma_1\gamma_2 + \gamma_2\gamma_3 + \gamma_1\gamma_3)}\left[\gamma_1\gamma_2\left\{\beta^2\left(\frac{\eta_1-\eta_3}{\gamma_1}\right) + \beta^2\left(\frac{\eta_2-\eta_3}{\gamma_2}\right) + \alpha\eta_3 - \beta\delta\left(I_{L_1} + I_{L_2} + I_{L_3}\right)\right\}\right] \tag{25}$$

$$\frac{\hat{v}_o(s)}{\hat{d}(s)} = \frac{\gamma_1\gamma_2\gamma_3}{\alpha\gamma_1\gamma_2\gamma_3 - \beta^2(\gamma_1\gamma_2 + \gamma_1\gamma_3 + \gamma_2\gamma_3)}\left\{(\alpha\phi - \beta\delta)\left(\frac{\eta_1}{\gamma_1} + \frac{\eta_2}{\gamma_2} + \frac{\eta_3}{\gamma_3}\right) + \left(\delta - \frac{\alpha\phi}{1-D}\right)\left(I_{L_1} + I_{L_2} + I_{L_3}\right)\right\} \tag{26}$$

$$\frac{V_o}{V_{in}} = \frac{NR(R+r_c)(1-D)}{ND^2R^2 - D[2NR^2 + R(r_d + r_c + r_s) + r_c(r_d + r_s)] + R(r_d + r_c + r_l) + r_c(r_d + r_l) + NR^2} \tag{27}$$

For equal parameters and neglecting the internal resistance of diode, switch and capacitor

$$\frac{\hat{v}_o(s)}{\hat{d}(s)} = \frac{\frac{3}{LC}\left[(1-D)V_c - I_L(sL + r_L)\right]}{s^2 + s\left(\frac{1}{RC} + \frac{r_L}{L}\right) + \frac{1}{LC}\left(3(1-D)^2 + \frac{r_L}{L}\right)} \quad (28)$$

and after neglecting E.S.R. of inductor

$$\frac{\hat{i}_L(s)}{\hat{d}(s)} = \frac{\frac{1}{LC}\left[sC(V_c + r_c I_L) + 3(1-D)I_L + \frac{1}{R}(V_c + I_L r_c)\right]}{s^2 + s\left(\frac{1}{RC} + \frac{(1-D)r_c}{L}\right) + \frac{(1-D)}{RLC}\left(r_c + 3(1-D)R\right)} \quad (29)$$

Also, from steady state analysis, we may obtain nominal parameters

$$[X_{ss}] = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \\ V_o \end{bmatrix} = -[A]^{-1}[B]V_{in}$$

IV. Stability Analysis

A non ideal circuit diagram for 3-phase interleaved boost converter is given in Fig. 2. The 3-phases are modeled by connecting three inductors in parallel with a common output capacitor, C_f across the output.

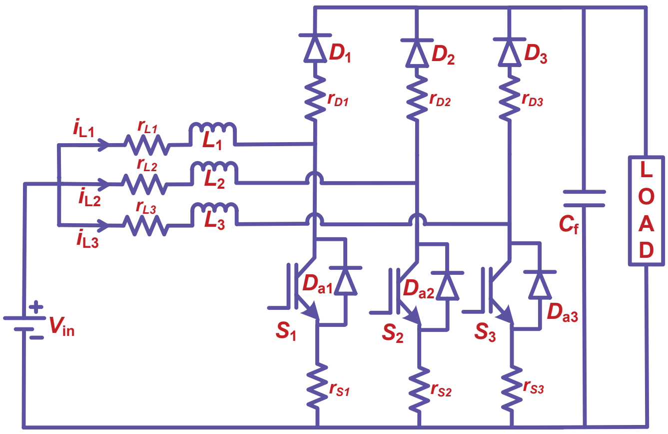

Fig. 2. 3-phase non ideal interleaved boost converter.

The specifications for a 3 phase interleaved boost converter are tabulated in Table I. Using (28) and (29), we get the control to output voltage transfer function as

$$\frac{\hat{v}_o(s)}{\hat{d}(s)} = \frac{-33264\ s + 1.799\times10^8}{s^2 + 2.83\ s + 3.815\times10^5} \quad (30)$$

and control to inductor current transfer function as

$$\frac{\hat{i}_L(s)}{\hat{d}(s)} = \frac{1.153\times10^4\ s + 1.153\times10^6}{s^2 + 50.17s + 2.704\times10^5} \quad (31)$$

TABLE I
SYSTEM PARAMETERS FOR STABILITY ANALYSIS

Parameters	***Value[Unit]***
Input voltage, V_{in}	180[volts]
Output Capacitance, C_f	100 [µF]
Inductor inductance, L	30 [mH]
Load resistance, R	200[Ω]
Duty ratio, d	0.48
Capacitor E.S.R. ,r_c	0.01[Ω]
Inductor E.S.R , r_l	0.01[Ω]
MOSFET on resistance, r_s	0.18[Ω]
Diode resistance, r_d	0.01[Ω]

Bode plots for uncompensated system with the control to output voltage transfer function shows a negative gain and phase margin in Fig. 3, implying an unstable system.

Formulating a lead lag compensator of the form $H_1(s)\times H_2(s)$, where $H_1(s)$ is given as

$$H_1(s) = K_1 \frac{1 + \frac{s}{\omega_{z1}}}{1 + \frac{s}{\omega_{p1}}}$$

and $H_2(s)$ should be

$$H_2(s) = K_1 \frac{1 + \frac{s}{\omega_{z2}}}{\frac{s}{\omega_{z2}}}$$

Developing the above mentioned compensator for the transfer function given below

$$G(s) = \frac{\left(1 - \frac{s}{\omega_z}\right)}{1 + \frac{s}{Q\omega_o} + \frac{s^2}{\omega_o^2}} \quad (32)$$

The Equ. (30) can be represented in the same form as (32) by

$$\frac{\hat{v}_o(s)}{\hat{d}(s)} = (471.55)\frac{\left(1 - \frac{s}{5408.0319}\right)}{1 + \frac{s}{(218.253)(617.6569)} + \frac{s^2}{(617.6569)^2}} \quad (33)$$

Comparing (33) with (32), we obtain

$\omega_o = 3985$ rad/sec

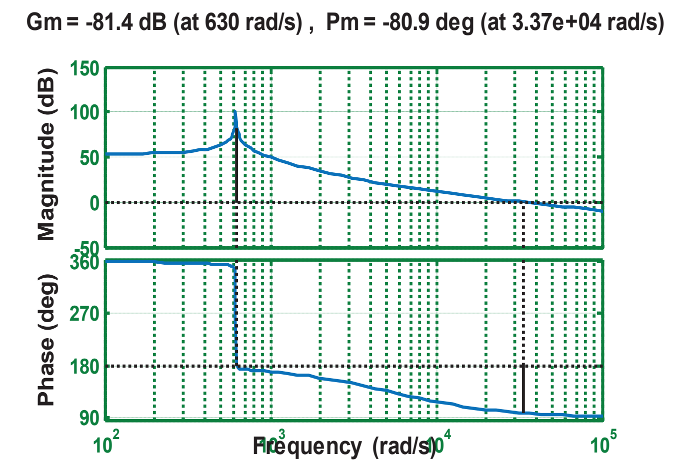

Fig. 3- Frequency response for control to output voltage transfer function (uncompensated)

Using the information about gain margin and phase margin from the bode plot of uncompensated system in Fig. 3, we formulate

$$H_1(s) = \frac{0.021s - 19852}{0.025s + 19852} \quad (34)$$

And

$$H_2(s) = \frac{s + 1870}{0.8*s} \quad (35)$$

So,

$$H_1(s)*H_2(s) = \frac{0.21s^2 - 19460s - 3.712\times10^7}{0.2s^2 + 35880s} \quad (36)$$

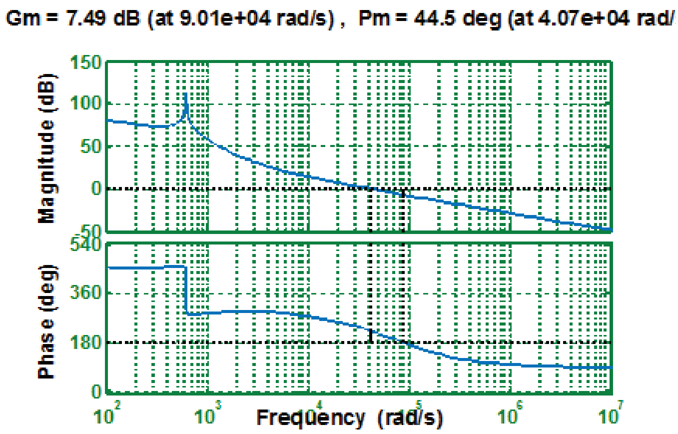

Fig. 4-Frequency response for control to output voltage transfer function with compensator

Using the lead lag compensator, we get the open loop frequency response as shown in Fig. 4. The system becomes stable after proper compensator design) with gain margin of 7.5 db and phase margin of 44.5 deg. Similarly, Fig.5 shows the frequency response for control to inductor current transfer function as derived in (31), without applying any compensation technique. The gain margin of the nominal plant comes out to be infinity whereas the phase margin equals to 85.1 deg. Hence, inherent stability is the distinct feature of this transfer function as the gain margin is infinite.

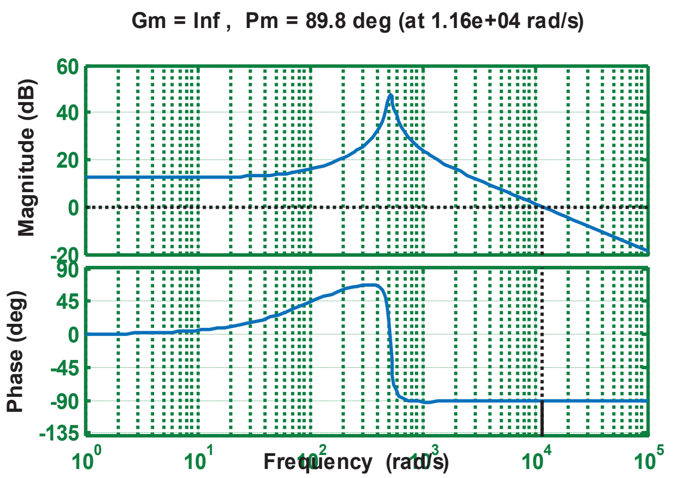

Fig. 5-Frequency response for control to inductor current transfer function (uncompensated)

For a 3 phase IBC, substituting N =3 in (27) the voltage gain becomes

$$\frac{V_o}{V_{in}} = \frac{3R(R+r_c)(1-D)}{3D^2R^2 - D[6R^2 + R(r_d + r_c + r_s) + r_c(r_d + r_s)] + R(r_d + r_c + r_l) + r_c(r_d + r_l) + 3R^2}$$

Neglecting the parasitic resistances, we obtain the gain for ideal IBC which is independent of the number of levels employed in the topology.

$$\frac{V_o}{V_{in}} = \frac{1}{(1-D)}$$

Fig. 6 shows the variation of voltage gain with duty ratio. Although the voltage gain differs slightly for d>0.4, such variation becomes critical for applications with low tolerance in voltage and require precise calculation of voltage gain by considering the actual model to be encountered in practical scenarios.

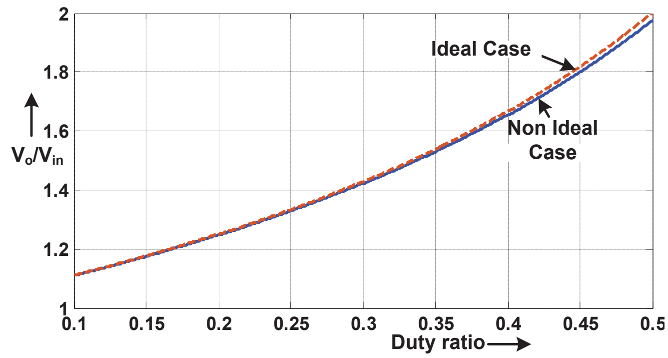

Fig. 6-Vo/Vin with duty ratio variation (ideal and non ideal cases)

V. DISCUSSION

The frequency analysis shows that the model is unstable in open loop when control to output voltage transfer function is considered. The compensator is designed and by testing the frequency according to the given relation stabilizes the system. The control to inductor current transfer function has infinite gain margin indicating inherent stability and as such the system is left uncompensated. The duty ratio relation indicates that IBC follow the same gain as of that of a conventional boost converter but for non ideal cases the gain decreases. There is a partial increase in gain for a 3 phase IBC and lesser current in each inductor branch implying a reduction in device stress.

VI. Conclusions

In this paper, the small signal analysis was carried for a generalized interleaved boost converter. The equations were developed by state space averaging and then perturbation was applied. The derived mathematical equations are obtained by considering all the parasitic and this feature is beneficial in designing a robust controller. The stability analysis can be carried out for any no. of phases for a given IBC, the same way it is shown for 3-level IBC .Stability analysis is performed for output voltage and inductor current with respect to control, which in this case is the duty cycle. Analysis has been shown by designing the compensator for open loop transfer function of output voltage to duty ratio. The transfer function models are second order equations irrespective of the no. of phases. The analysis for an interleaved boost converter becomes cumbersome with increase in the no. of phases. However with the given equations the transfer function equations can be obtained directly.

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